

RUSK COUNTY AP, WI, USA

WMO: 723758

Lat: 45.497N

Lon: 91.001W

Elev: 377

StdP: 96.87

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 54928

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-26.5	-23.4	-30.4	0.2	-25.3	-27.6	0.3	-22.6	10.1	-10.5	8.9	-8.8	1.7	300	0.553

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	30.0	22.1	28.4	21.0	27.2	20.0	23.6	27.9	22.6	26.6	21.6	25.3	4.3	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.5	18.0	25.8	21.3	16.7	24.8	20.3	15.7	23.9	72.6	28.0	68.5	26.8	64.7	25.5	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 8.9	7.9	7.1	-31.1	32.8	2.2	1.7	-32.6	34.0	-33.9	35.0	-35.1	36.0	-36.6	37.3		
(5)			-31.2	25.5	2.1	1.3	-32.8	26.4	-34.0	27.2	-35.2	27.9	-36.7	28.9	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	6.4	-9.7	-8.8	-0.7	6.1	13.2	18.1	20.6	19.1	15.3	8.1	0.8	-6.6	(6)	
	DBStd	11.94	7.36	6.91	6.95	5.32	4.79	3.39	3.05	2.94	4.49	4.87	5.52	6.38	(7)	
	HDD10.0	2533	609	528	338	137	21	1	0	0	8	97	280	516	(8)	
	HDD18.3	4572	868	761	589	366	172	44	14	26	113	319	527	774	(9)	
	CDD10.0	1207	0	0	7	21	119	243	328	283	166	38	3	0	(10)	
	CDD18.3	207	0	0	1	0	12	37	83	51	21	2	0	0	(11)	
	CDH23.3	1778	0	0	1	6	138	327	739	395	158	13	0	0	(12)	
	CDH26.7	388	0	0	0	0	32	63	191	73	29	1	0	0	(13)	
(14) Wind	WSAvg	3.0	3.1	3.1	3.1	3.6	3.2	2.8	2.4	2.3	2.7	3.1	3.2	3.0	(14)	
(15) Precipitation	PrecAvg	832	27	23	46	73	89	119	98	110	96	72	48	30	(15)	
	PrecMax	1096	91	69	102	163	164	200	190	258	209	154	177	50	(16)	
	PrecMin	641	5	3	13	15	41	54	41	18	21	22	1	8	(17)	
	PrecStd	129	19	15	24	35	38	44	41	52	46	38	38	12	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	5.2	9.6	21.3	24.0	30.2	30.6	32.7	30.8	29.9	25.2	17.8	7.8	(19)	
		MCWB	2.7	6.4	15.3	14.4	20.6	22.9	24.4	22.6	22.2	18.0	12.0	5.2	(20)	
	2%	DB	2.8	4.9	16.1	21.0	27.0	28.3	30.2	28.6	27.3	22.2	14.6	4.9	(21)	
		MCWB	1.8	2.7	11.5	12.4	18.1	20.5	22.5	21.8	20.5	16.9	10.5	3.6	(22)	
	5%	DB	1.5	2.4	12.1	18.0	24.6	26.8	28.7	27.2	25.1	19.5	11.2	2.6	(23)	
		MCWB	0.8	0.7	8.1	10.7	16.8	19.2	21.6	20.1	19.2	14.6	8.2	1.6	(24)	
	10%	DB	0.3	0.4	8.2	15.5	22.1	25.2	27.3	25.6	23.0	16.2	8.6	1.4	(25)	
		MCWB	-0.4	-0.8	5.3	9.3	15.6	18.1	20.3	19.5	18.2	12.6	6.2	0.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.1	6.3	15.8	15.9	23.0	24.3	26.0	24.6	23.4	21.7	13.9	6.5	(27)	
		MCDB	4.2	8.9	20.0	20.2	28.7	28.7	30.6	28.4	27.9	23.6	16.1	7.1	(28)	
	2%	WB	2.0	2.9	12.7	13.7	20.3	22.2	23.8	23.1	21.9	17.3	10.8	3.5	(29)	
		MCDB	2.7	4.4	14.7	19.4	24.0	27.0	28.5	26.5	25.1	20.8	13.9	4.5	(30)	
	5%	WB	1.0	0.8	8.2	11.8	18.3	20.6	22.7	22.1	20.7	15.2	8.3	1.9	(31)	
		MCDB	1.6	2.1	11.2	16.6	22.4	24.6	26.9	25.6	23.6	18.9	10.5	2.5	(32)	
	10%	WB	-0.2	-0.6	5.5	9.9	16.6	19.5	21.6	20.9	19.3	12.7	6.4	0.8	(33)	
		MCDB	0.3	0.3	8.1	15.1	20.5	23.4	25.6	24.0	22.5	15.7	8.6	1.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.0	10.7	10.3	10.8	12.2	11.3	11.5	11.3	11.4	9.5	7.8	7.4	(35)	
		MCDBR	8.1	10.0	13.1	15.9	14.8	13.0	12.9	12.8	13.0	13.7	12.3	7.5	(36)	
	5% WB	MCWBR	7.1	8.1	8.2	8.8	7.3	6.4	6.2	6.4	6.9	8.3	8.6	6.3	(37)	
		MCWBR	7.4	8.8	11.2	13.9	12.1	10.9	11.2	10.6	10.5	11.6	11.5	6.8	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.248	0.268	0.301	0.344	0.374	0.375	0.372	0.376	0.352	0.312	0.284	0.257	(40)	
		taud	2.391	2.334	2.387	2.396	2.346	2.363	2.395	2.396	2.453	2.550	2.527	2.441	(41)	
	Ebn at Noon		882	922	934	916	894	892	889	873	867	864	833	832	(42)	
		Edh at Noon	71	92	102	111	121	120	114	110	96	75	63	62	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.63	2.74	3.88	4.56	5.31	5.85	6.09	5.11	3.98	2.50	1.51	1.12	(44)	
	RadStd		0.18	0.22	0.36	0.47	0.40	0.46	0.26	0.36	0.25	0.28	0.15	0.13	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (50 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page